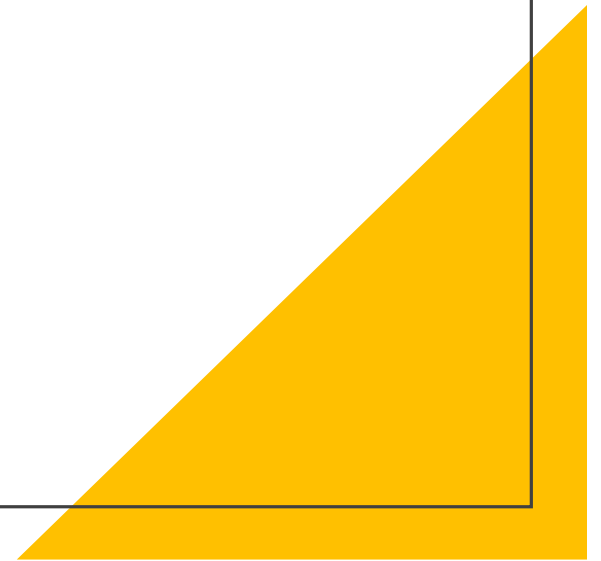




Final Study Guide

Advanced Principles Fall 2025

Lab 5



Intracranial Mass Lesion

The location of mass lesion determines symptoms

- **Primary motor cortex:** contralateral weakness
- **Blood vessels** (distortion, irritation): headache
- **Ventricular compression** (obstruction of CSF flow, hydrocephalus): Nausea and vomiting, blurred / double vision and/or unstable balance, poor coordination
- **Displacement/stretching of upper brainstem:** Impaired consciousness, coma

Hydrocephalus

“Water in the head”: Excess CSF in the intracranial cavity

- Possible Causes:
 - Excess CSF production
 - Obstruction of flow at any point
 - Decreased reabsorption via arachnoid granules
- Communicating hydrocephalus: deficit in CSF reabsorption, Impairment of arachnoid granulations
- Noncommunicating hydrocephalus: CSF flow obstruction

Symptoms: headache, nausea, vomiting, cognitive impairments, decreased consciousness, decreased vision

Brain Tumors

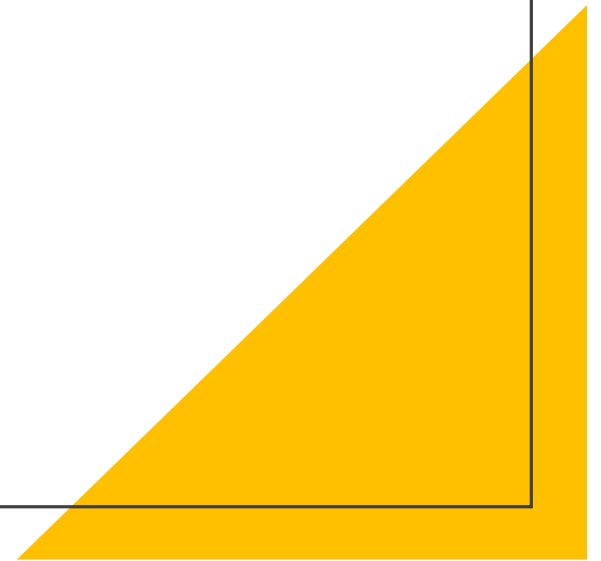
Categories:

- Primary: tumor that starts in the brain
- Metastatic: begins as a tumor in another part of the body

Magnetic Resonance Imaging (MRI)

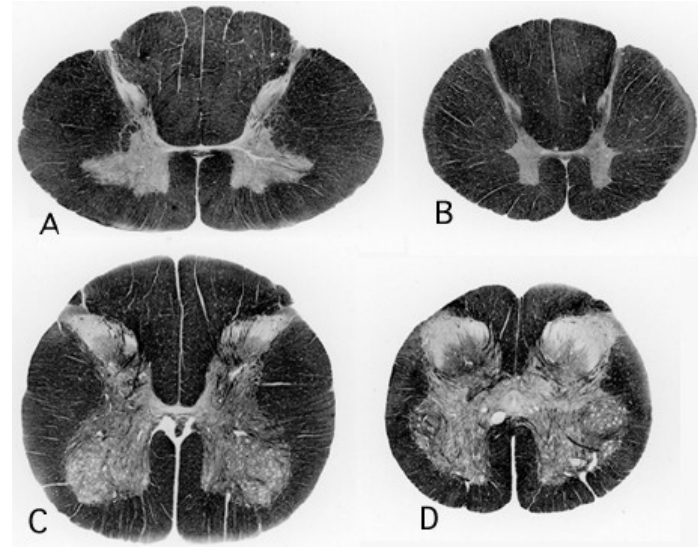
- T1-weighted image: measures the realignment of original magnetic field following RF pulse (“righting” component)
- T2-weighted image: measures the protons falling out of synchrony (“dephasing” component)
- Proton (spin) density weighted image: measures number of magnetizable protons per volume

Lab 6



Spinal Cord Segments

- A – Cervical
- B – Thoracic
- C – Lumbar
- D - Sacral



Types of Tracts

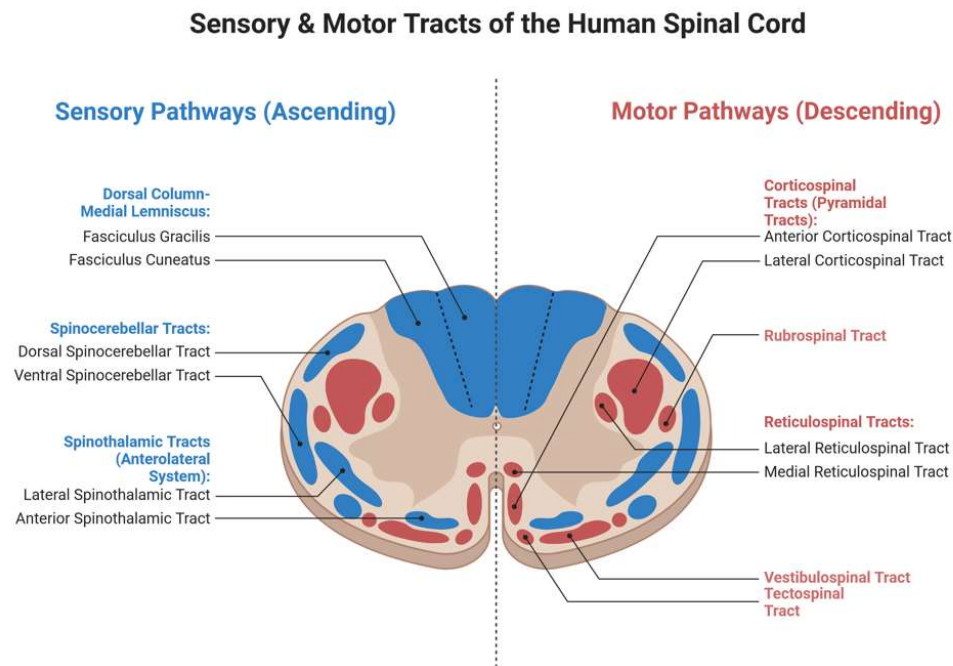
↑ Ascending Spinal Tracts

- Periphery to spinal cord and brain

↓ Descending Spinal Tracts

- Cortex to periphery and exterior levels

Make sure you know the location of each of these tracts!



Tract	Origin	Decussation	Location in Cord	Terminus	Function
Posterior (Dorsal) Columns					
Gracile fasciculus ^a	Dorsal root ganglion	Medulla (internal arcuate)	Posterior column (medial)	Thalamic ventral posterolateral nucleus	Fine touch Vibration Proprioception (lower body)
Cuneate fasciculus ^a	Dorsal root ganglion	Medulla (internal arcuate)	Posterior column (lateral)	Thalamic ventral posterolateral nucleus	Fine touch Vibration Proprioception (upper body)

Corticospinal Tracts

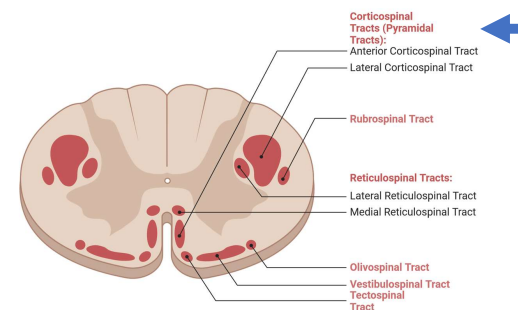
	Lateral	Ventral
Origin	Primary Motor Cortex	Primary Motor Cortex
Decussation	Pyramids (medulla)	none
Termination	Entire spinal cord	Cervical & Upper Thoracic
Function	Movement of contralateral limbs	Control of bilateral axial and girdle muscles

Voluntary, skilled movement

Decussation at pyramids
(medulla)

- 80-85 percent cross-over

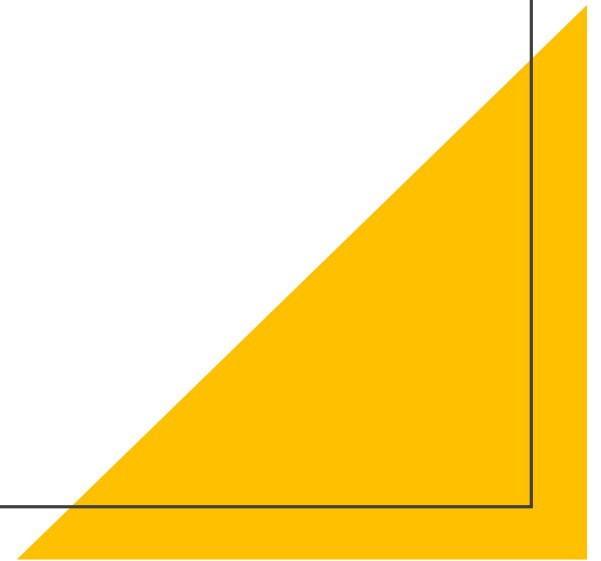
Motor Tracks and Pathways (Descending) of the Human Spinal Cord



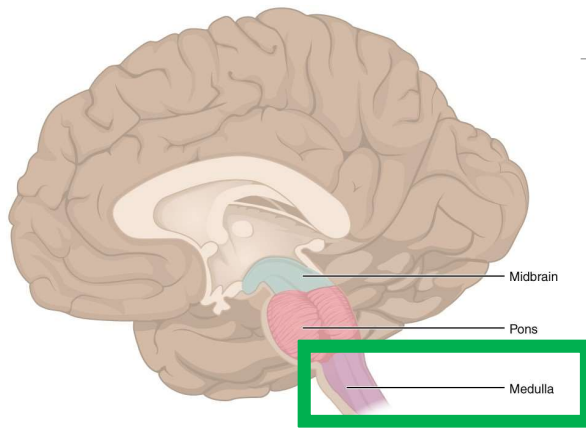
Other Motor Tracts

	Rubrospinal	Lateral Vestibulospinal	Medial Vestibulospinal	Pontine Reticulospinal	Medullary Reticulospinal
Origin	Red nucleus (midbrain)	Vestibular nuclei	Vestibular nuclei	Reticular formation	Reticular formation
Decussation	Ventral tegmentum (midbrain)	None	None	None	None
Termination	Cervical cord	Entire cord	Cervical & Upper Thoracic	Entire cord	Entire cord
Function	Muscle tone	Maintaining balance	positioning of head and neck	Automatic posture and gait-related movements ; axial and limb muscles	Automatic posture and gait-related movements

Lab 7



Medulla Function



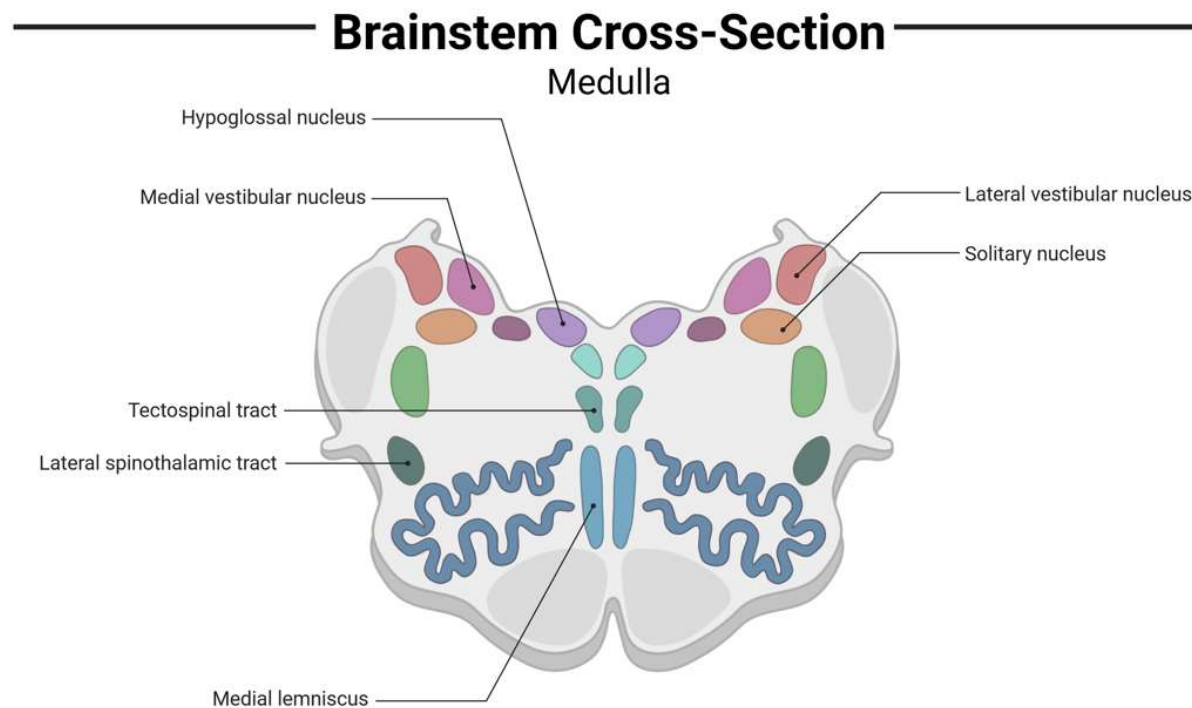
Autonomic Control

- Cardiac – sympathetic and parasympathetic functioning
- Respiratory – chemoreceptors: **detect variation of the oxygen concentration in blood**
- Reflexes – vomiting, coughing, sneezing, and swallowing
- Vasomotor – baroreceptors: detect changes in blood pressure

Decussation of sensory fibers

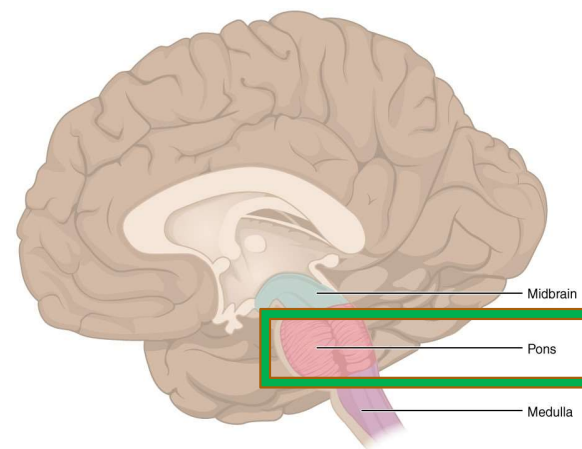
- Obex

Make sure you know the location of each of these tracts!



Pons Function

- Autonomic nerve pathways
- Integration of information
- Exerts some control over:
 - Regulation of respiration
 - involuntary actions
 - Sensory roles: hearing, equilibrium, taste, and in facial sensations such as touch and pain
 - Motor roles: eye movement, facial expressions, chewing, swallowing, and the secretion of saliva and tears



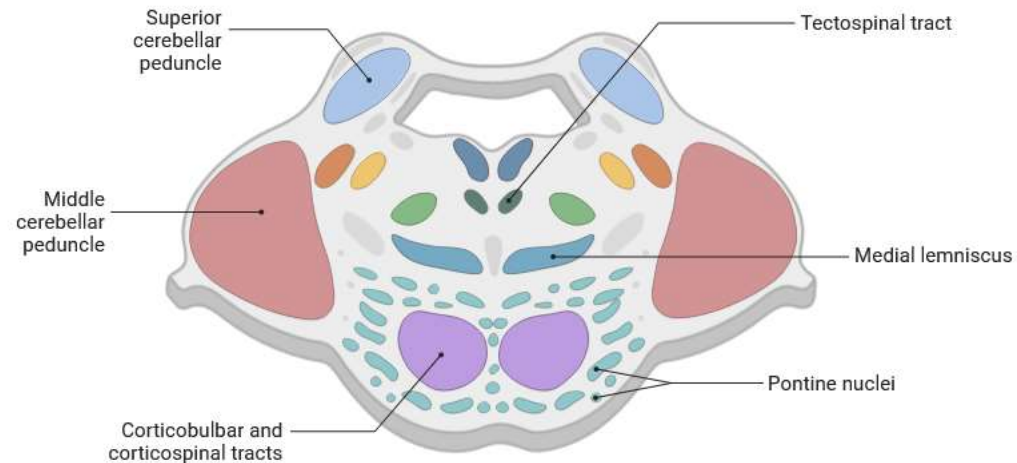
Pons

Structures

- 2 major divisions:
 - Basilar part of the pons (ventral pons)
 - Pontine tegmentum (dorsal pons)
- Lemniscus = secondary sensory fiber
- Fasciculus = nerve and motor bundles

Make sure you know the location of each of these tracts!

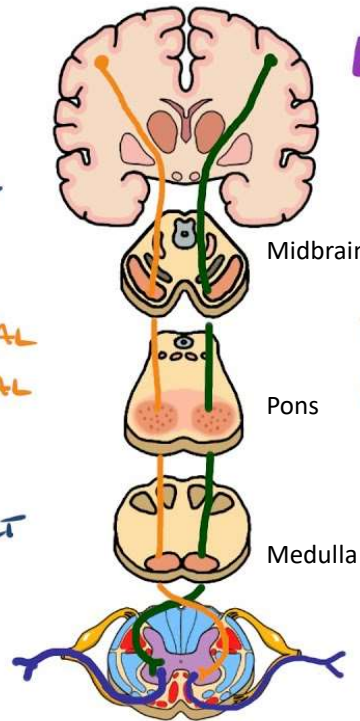
Brainstem Cross-Section
Pons



PYRAMIDAL

- ① **CORTICOSPINAL TRACT**
VOLUNTARY CONTROL
- BODY
LATERAL CORTICOSPINAL
ANTERIOR CORTICOSPINAL

- ② **CORTICOBULBAR TRACT**
VOLUNTARY CONTROL
- FACE, HEAD, NECK



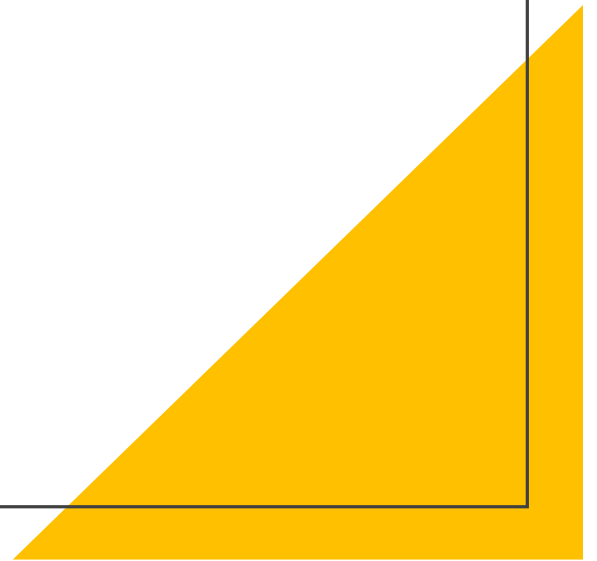
EXTRAPYRAMIDAL

- 1) **RUBROSPINAL**
FINE MOTOR CONTROL
- 2) **RETICULOSPINAL**
MEDIAL - CONTRACTION INCREASES TONE
LATERAL - RELAXATION DECREASES TONE
- 3) **VESTIBULOSPINAL**
BALANCE AND POSTURE
- 4) **TECTOSPINAL / COLLICULOSPINAL**
HEAD CO-ORDINATION

Tract	Path	Function
corticospinal tract	Motor Cortex → Spinal Cord (LMN in ventral horn)	voluntary motor control of the limbs and voluntary posture adjustments
corticobulbar tract	Motor Cortex → Medulla and Pons	swallowing, tongue movement, facial and jaw muscles
colliculospinal tract (tectospinal tract)	Superior Colliculus → LMN	involuntary head movements
rubrospinal tract	Red Nucleus → LMN	involuntary arm movements (for balance) and body support
vestibulospinal tract	Vestibular Nucleus → LMN	posture and balance
reticulospinal tract	Reticular Formation → LMN	various involuntary motor movements and assists with balance

LMN = lower motor neurons

Lab 8

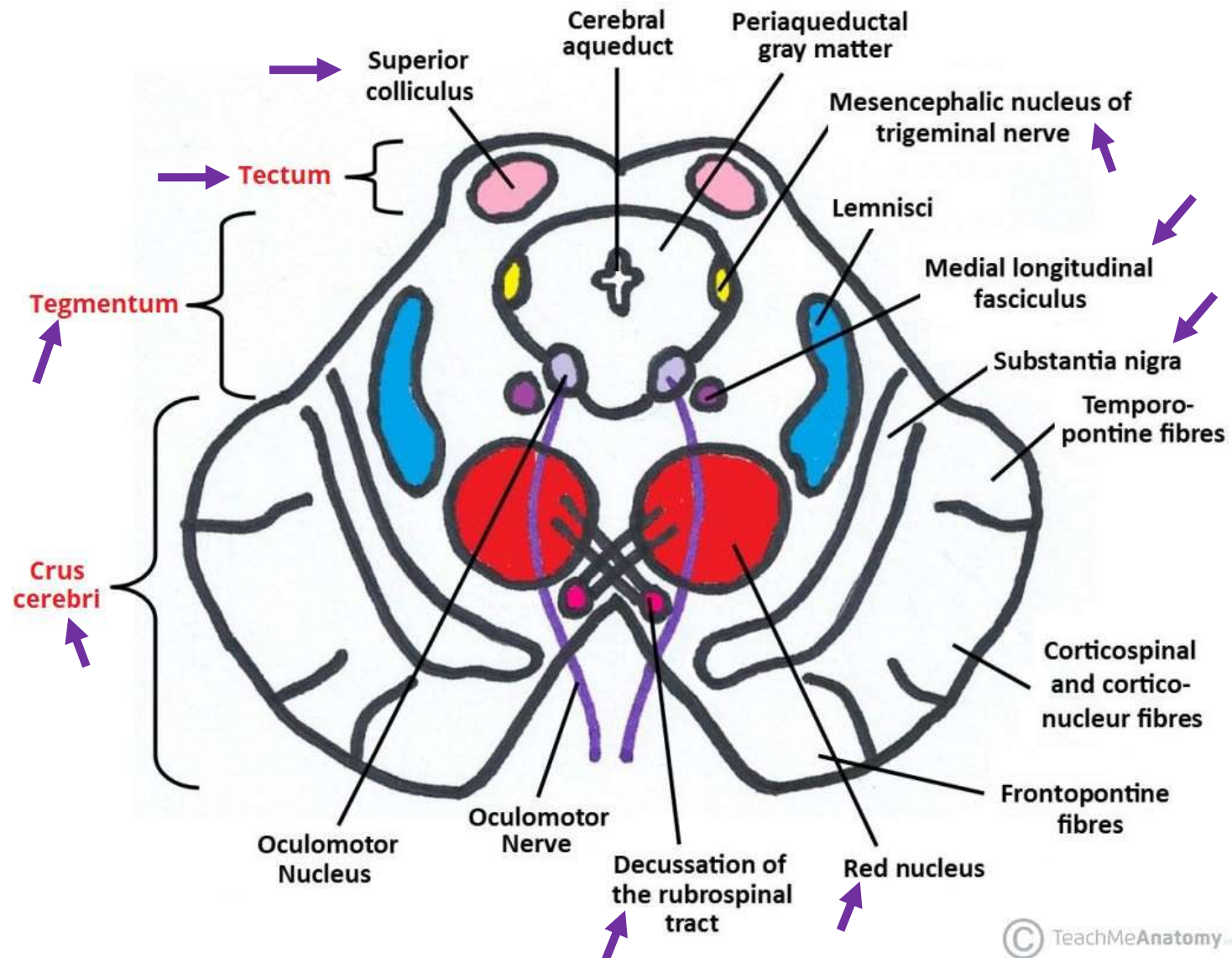


Midbrain Functions

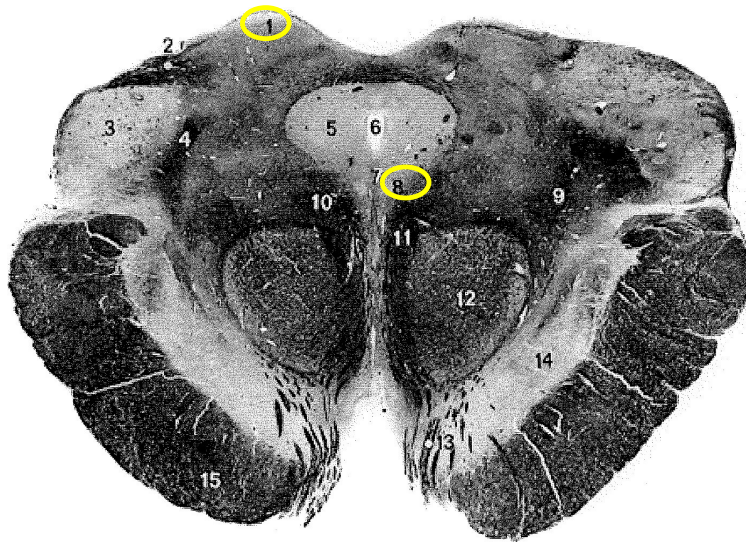
- Also known as the Mesencephalon
- Associated senses:
 - Vision
 - Hearing
- Associated autonomic functions
 - Motor control
 - Sleep/wake
 - Arousal (alertness)
 - Temperature regulation

Rostral Midbrain

Only worry about structures that are pointed out with an arrow

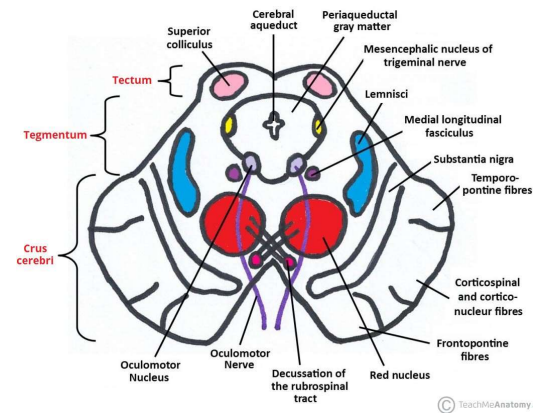


Same as previous slide, only worry about the structures outlined in the previous slide



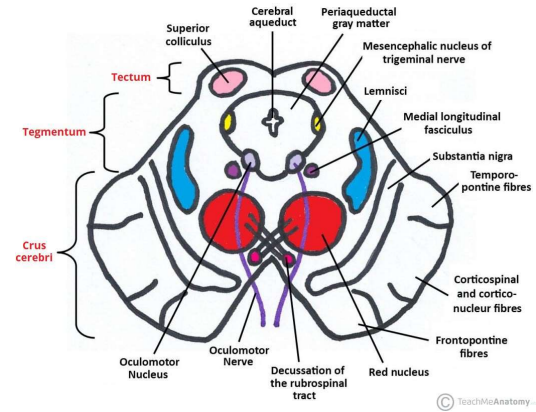
- 1. Superior colliculus
- 2. Brachium of superior colliculus
- 3. Medial geniculate nucleus
- 4. Brachium of inferior colliculus
- 5. Central gray substance
- 6. Cerebral aqueduct
- 7. Edinger-Westphal nucleus
- 8. Nucleus of oculomotor nerve
- 9. Medial lemniscus
- 10. Central tegmental tract
- 11. Medial longitudinal fasciculus
- 12. Red nucleus
- 13. Fibers of oculomotor nerve
- 14. Substantia nigra
- 15. Basis pedunculi

Midbrain Structures



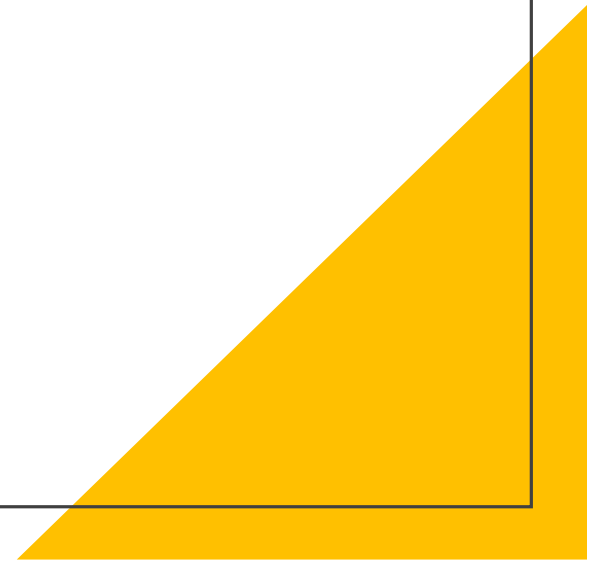
Structure	Function
The reticulospinal tract	exerts some control over alertness and takes input from the tectum
The superior colliculi	process visual information, aid the decussation of optic nerve fibers, and is involved in saccadic eye movements
The inferior colliculi	process certain auditory information

Tegmentum Structures



Structure	Function
Red Nuclei	Gait
Medial Lemniscus	Somatosensation from the skin and joints
Reticular Formation	Production of serotonin, Arousal, Consciousness, Motor coordination, Exhalation

Lab 9



12 Cranial Nerves

- 1-olfactory nerve
- 2-optic nerve
- 3-oculomotor nerve
- 4-trochlear nerve
- 5-trigeminal nerve
- 6-abducens nerve
- 7-facial nerve
- 8-vestibulocochlear nerve
- 9-glossopharyngeal nerve
- 10-vagus nerve
- 11-accessory nerve
- 12-hypoglossal nerve

Nerve	Number	Somatic Motor	Branchial Motor	Visceral Motor	Visceral Sensory	General Sensory	Special Sensory	Function
Olfactory	1						X	Smell
Optic	2						X	Vision
Oculomotor	3	X						Innervates all the extrinsic eye muscles except superior oblique and lateral rectus
				X				PSNS supply to ciliary and pupillary constrictor muscles
Trochlear	4	X						Motor to superior oblique
Trigeminal	5		X					Motor to muscles of mastication
						X		Somatic sensation from face
Abducens	6	X						Motor to lateral rectus
Facial	7		X					Innervate muscles of facial expression
				X				PSNS supply to glands of head
						X		General sensation from external ear tympanic membrane
							X	Taste, anterior 2/3 of tongue
Vestibulocochlear	8						X	Impulses from inner ear; equilibrium and hearing
Glossopharyngeal	9		X					Motor to stylopharyngeus muscle (swallowing)
				X				PSNS supply to parotid (salivary) gland
					X			Baroreceptors of the carotid sinus and chemoreceptors of the carotid body
						X		Posterior 1/3 of tongue, tympanic membrane
							X	Taste, posterior 1/3 of tongue
Vagus	10		X					Pharynx and larynx
				X				PSNS supply to laryngeal and pharyngeal muscles and abdominal viscera
					X			Enable a vasoconstrictor response and increase blood pressure
						X		Sensation from a small area around the external ear
Accessory	11		X					Motor to sternomastoid and trapezius muscle
Hypoglossal	12	X						Swallowing and speech articulation